

Cooperative Multi-Agent Reinforcement Learning for EV Charging Station Selection and Grid Stability

Problem Statement

Modern urban power grids are entering an Electric Vehicle (EV) era where charging demand is never static. On any given day:

- Large numbers of EVs converge on the same nearby charging stations at similar times (commute hours, evenings)
- Individual driver navigation choices are made independently, with no visibility into what other drivers are simultaneously choosing
- Local transformers and feeders have hard physical load limits that do not scale with demand
- Electricity spot prices and grid stress vary by time of day and by location
- Charging station occupancy changes continuously and unpredictably as vehicles arrive and depart

Despite this volatility, most real-world EV navigation systems still route drivers using simple heuristics:

- Nearest-station routing always send the driver to the closest available charger
- Shortest-wait routing send the driver to whichever station currently reports the shortest queue
- Static/tiered pricing station prices are fixed or change only on a fixed schedule, not in response to real-time grid stress

These rules are simple to implement, but they do not reason about the collective effect of many drivers making the same locally-optimal choice at once, and they do not adapt to combinations of disruptions.

Consequence when things go wrong: when many EVs independently choose the same "nearest" station during peak hours, the result is long queues for drivers and localized transformer overload for the grid even though other stations nearby may be sitting idle. Rule-based systems cannot resolve this because no single station or driver has visibility into system-wide demand.

The problem this project addresses is to design a cooperative decision-making framework in which EVs and charging stations are both intelligent, learning agents that continuously adapt the EV agent choosing which station to head to, and the station agent adjusting its real-time price so that driver wait time, travel distance, and grid load stress are jointly minimized, rather than optimizing one at the expense of the other.

Base Paper

The paper titled "Cooperative Multi-Agent Reinforcement Learning for Grid-Aware EV Charging Management with Cross-Site Redirection" (Yang & Tang, 2025) addresses the problem of EV charging congestion across multiple sites in a shared grid region. Charging sites are modelled as cooperative agents using the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm, trained in a spatiotemporal environment that integrates real public charging records with simulated domestic demand from the Perth and Kinross region of Scotland.

The proposed solution works by training a cross-site actor under Centralized Training, Decentralized Execution (CTDE): during training, a shared critic observes the state of all charging sites simultaneously, allowing sites to learn to redirect EVs toward less-congested locations before congestion peaks arrive. A delayed replay buffer with short-horizon forecasts allows redirection decisions to anticipate near-future congestion rather than only reacting to current occupancy. When tested against a baseline of uncoordinated, independent site operation, the learned policy reduced peak-hour load standard deviation by up to 40% and lowered cumulative threshold violations by 37%.

However, the method treats only the charging stations as learning agents that redirect demand; individual EVs are not modelled as independent decision-makers with their own state (battery level, position) and reward. Coordination is achieved purely through site-to-site redirection signals, not through a market-like pricing mechanism that both sides respond to, and grid constraints are represented as load thresholds rather than a full electrical network model which is the gap this project aims to address.

Literature Review

Reinforcement learning has emerged as a promising alternative to static, rule-based dispatch for EV charging management, with several notable contributions in recent years.

Park & Moon (2022) [1] proposed a multi-agent deep reinforcement learning approach with centralized training and decentralized execution to derive a charging/discharging schedule for each EV visiting a smart-grid-connected charging station, using the COMA and MADDPG algorithms. The approach focuses on minimizing individual EV operating cost and does not address cross-station driver routing or station-side dynamic pricing.

Yang & Tang (2025) [2] the base paper modelled charging sites as cooperative MADDPG agents performing cross-site EV redirection under a CTDE framework, achieving significant reductions in peak-load variance and threshold violations. The framework does not model EVs as independent learning agents, nor does it use pricing as a coordination signal.

A real-time online charging control study (ScienceDirect, 2025) [3] proposed an attention-network-based multi-agent deep reinforcement learning strategy for scheduling within a single charging station, using an action-mask layer to filter invalid actions and a CTDE framework to maximize both

operator revenue and driver comfort. This work operates within one station and does not address inter-station routing or grid-wide load balancing.

Li et al. (2024) [4] introduced a decentralized collaborative optimal scheduling framework across multiple charging stations with heterogeneous facility types, using multi-agent reinforcement learning to coordinate charging strategy. Grid load is considered at an aggregate level, without a physical network model of voltage or transformer constraints.

A decentralized collaborative pricing and shunting study [5] applied multi-agent reinforcement learning to let multiple charging stations jointly set prices and shunt (redirect) demand between each other, closely resembling the station-pricing component of this project, but without an independently-modelled EV routing agent responding to those prices as a separate learner.

A broader review of RL applications in EV charging station management [6] surveys value-based, actor-critic, and multi-agent approaches across dispatch, pricing, and demand-response tasks, and notes that most existing systems optimise from a single side of the problem either station scheduling or vehicle routing rather than treating both as jointly learning, cooperative agents.

Research Gap

Across the reviewed literature, existing multi-agent reinforcement learning systems for EV charging consistently address station-side scheduling, pricing, or redirection, but rarely model the EV itself as an independent agent that learns to respond to those signals. Table 1 summarizes this comparison.

Objective	Existing Works
Station-side load/occupancy awareness	Yes
Independent EV-side routing agent (own state & reward)	Partial / No
Dynamic pricing as a coordination mechanism	Partial
Joint driver-experience + grid-stability reward	No
Real electrical network model (voltage/transformer physics)	Partial
Real-time coordinated pricing + routing together	No

None of the reviewed approaches jointly models an independently-learning EV routing agent and a station-side dynamic pricing agent under a single shared multi-objective reward that also incorporates real electrical-grid physics. This absence of a combined, two-sided, grid-aware cooperative framework constitutes the research gap this work addresses.

Proposed Solution

The proposed solution, GridCharge-RL, is a cooperative multi-agent reinforcement learning framework in which EVs and charging stations are both independent learning agents, coordinated through a shared multi-objective reward. The framework is structured as follows.

1. Environment Simulation

A city-scale agent-based simulation is constructed using the Mesa framework, representing EVs moving through a virtual road network toward charging stations. Each station is linked to a corresponding node in an electrical distribution network modelled using pandapower and a standard IEEE test feeder (e.g., IEEE 33-bus), so that grid stress reflects realistic voltage and transformer-loading behaviour rather than an arbitrary capacity number.

2. Agent Representation

Agent	Observes (State)	Learns to Decide (Action)
EV Agent	GPS position, battery state of charge, distance to nearby stations, current price at each station	Which charging station to navigate toward
Station Agent	Occupied charging bays, current local grid load/capacity, current electricity spot price	Local pricing tariff multiplier (Low / Medium / High)

3. Reinforcement Learning Formulation (State-Action-Reward)

The problem is formulated as a Markov Decision Process and solved using two coordinated algorithms:

- Environment: the Mesa + pandapower co-simulation described above, which evolves EV positions, station occupancy, and grid load forward in time.
- State (st): the local observation of each agent, as summarized in the table above, encoded as a feature vector per agent.
- Action (at): EV agents select a target station (discrete choice); station agents select a pricing multiplier (continuous/discrete adjustment).
- Reward (rt): a shared, weighted composite signal: $R = -(w_1 \cdot \text{Driver Wait Time}) - (w_2 \cdot \text{Distance Traveled}) - (w_3 \cdot \text{Grid Peak Load Stress})$, so no agent can improve its own outcome at the system's expense.
- Episode termination: an episode ends after a fixed simulated time horizon (e.g., a 24-hour cycle) or once all active EV charging requests in that horizon are resolved.

4. Algorithms

- EV Agent: Proximal Policy Optimization (PPO) well suited to the discrete station-selection action and stable to train in a non-stationary multi-agent setting.
- Station Agent: Multi-Agent Deep Deterministic Policy Gradient (MADDPG) designed for continuous actions (pricing) and multiple simultaneously-learning agents.
- Training paradigm: Centralized Training, Decentralized Execution (CTDE) a shared critic observes global state during training so agents learn to account for one another; at execution, each agent acts only on its own local observation.

5. Disruption Handling

Because each agent re-observes its local state and re-decides at every simulation step rather than following a pre-computed fixed plan, the framework naturally adapts to disruptions such as a station suddenly filling up, a surge of new charging demand, or a temporary drop in grid capacity, without requiring a full re-optimization from scratch.

6. Training and Evaluation Protocol

The agents are trained and evaluated within the simulated environment across multiple 24-hour demand cycles, with weekday and weekend demand patterns modelled separately. Performance is compared against a nearest-station greedy baseline (no learning, no pricing coordination), using consistent metrics: average driver wait time, average distance travelled, peak-hour grid load standard deviation, and cumulative grid threshold violations the same metrics used in the base paper, enabling direct comparison.

Expected Outcomes

- A working simulation environment modelling EV mobility, station occupancy, dynamic pricing, and realistic grid load (via pandapower) within a unified framework.
- A trained PPO-based EV routing agent and a MADDPG-based station pricing agent, coordinated via CTDE, capable of making real-time decisions as conditions change.
- Demonstrated improvement over a nearest-station greedy baseline across driver wait time, distance travelled, and grid peak-load metrics.
- Demonstrated improvement, or at minimum competitive performance, relative to the base paper's station-only redirection approach, evaluated under the added complexity of an independently-learning EV-side agent.
- A generalizable two-sided cooperative policy capable of extending to additional stations or a larger simulated city without retraining from scratch.
- A quantitative analysis of the trade-off between driver-side objectives (wait time, distance) and grid-side objectives (load stability), showing how the learned joint policy balances both under varying demand conditions.
- A foundation suitable for future extension toward real-world piloting via OCPP-integrated charging networks and utility grid-data partnerships.

References

- [1] K. Park and I. Moon, "Multi-agent deep reinforcement learning approach for EV charging scheduling in a smart grid," *Applied Energy*, vol. 328, p. 120111, 2022.
- [2] Y. Yang and R. Tang, "Cooperative multi-agent reinforcement learning for grid-aware EV charging management with cross-site redirection," 2025. [Base Paper]
- [3] "Real-time online charging control of electric vehicle charging station based on a multi-agent deep reinforcement learning," *ScienceDirect*, 2025.

[4] Y. Li et al., "Decentralized collaborative optimal scheduling for EV charging stations based on multi-agent reinforcement learning," IET Generation, Transmission & Distribution, 2024.

[5] "Decentralized collaborative pricing and shunting for multiple EV charging stations based on multi-agent reinforcement learning," arXiv:2406.11496.

[6] "Multi-agent deep reinforcement learning approach for EV charging scheduling in a smart grid" (review), ResearchGate/ScienceDirect, 2024.